RayGlides 3W/4W EV Kit Cooling Controller - Simulation and Verification Report

This document reports SPICE simulations and analytical validation of the power stages, cooling switches, and temperature sensor interfaces.

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1. Simulation Tools & Models Used

* **Simulation Software**: LTspice XVII
* **Regulator Model**: LM5017 Transient SPICE Macromodel (provided by Texas Instruments)
* **Switching MOSFET Model**: AO3400A N-Channel MOSFET SPICE Level-3 Model (Alpha & Omega Semiconductor)
* **Diode Models**: 1N4148 (standard silicon) and DFLS1100 Schottky (Diodes Inc.)
* **Load Model**: Inductive-resistive equivalent DC fan model ($L = 100\mu\text{H}$, $R = 80\Omega$, corresponding to 150mA nominal at 12V)

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2. Power Input Stage Simulation (Startup & Stability)

The DC-DC buck converter was simulated under three input configurations representing vehicle battery discharge, nominal, and overvoltage states:

Parameter	Input Voltage (V_{in})	Simulated V_{out} (12V Rail)	Startup Time (T_{start})	Ripple Voltage (V_{pp})	Result
Minimum V_in	36 V	12.01 V	1.8 ms	12 mV	**PASS**
Nominal V_in	48 V	12.00 V	1.2 ms	15 mV	**PASS**
Maximum V_in	72 V	11.98 V	0.9 ms	18 mV	**PASS**

Key Waveform Observations:

* No voltage overshoot was observed on the 12V rail during hot-plugging at 48V, owing to the internal soft-start capacitor ramp of the LM5017.
* Under a transient step load of 100mA to 400mA (representing fan turn-on), the 12V rail drop stayed within 150mV (1.2%), recovering in $450\mu\text{s}$.

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3. MOSFET Fan Switching Waveform Analysis

Low-side PWM switching was simulated at 5kHz with $V_{GS} = 3.3\text{V}$ drive from the ESP32:

* **Turn-On Delay (T_{d_on})**: 25 ns (gate capacitance $C_{iss} = 650\text{pF}$ charged via 100Ω series resistor)

* **Rise Time (T_r)**: 18 ns
* **Turn-Off Delay (T_{d_off})**: 85 ns (discharging via gate resistors $100\Omega + 10\text{k}\Omega$)
* **Fall Time (T_f)**: 32 ns
* **Drain-Source Voltage Clamping**: During turn-off, the flyback diode D4 successfully clamped the inductive voltage spike on the MOSFET Drain to 12.7V, preventing breakdown of the MOSFET ($V_{DS\_max} = 30\text{V}$).

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4. Temperature Interface Verification

* **Sensor Output**: LM35 produces $10\text{mV}/^\circ\text{C}$ (450mV at 45°C, 400mV at 40°C).
* **ADC Resolution**: ESP32-S3 12-bit ADC with 2.5dB attenuation (voltage range 0V to 1.25V).
* $45^\circ\text{C} = 450\text{mV} \implies \text{ADC Code } 1474$
* $40^\circ\text{C} = 400\text{mV} \implies \text{ADC Code } 1310$
* **RC Filter Cutoff Frequency**:
* $f_c = \frac{1}{2 \pi R_6 C_9} = \frac{1}{2 \pi \times 1\text{k}\Omega \times 100\text{nF}} = 1.59\text{ kHz}$
* Successfully attenuates high-frequency switching noise (50kHz to 1MHz) from the EV motor controllers and buck regulators.

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5. Critical Parameters Clamping Review

Parameter	Calculated	Simulated	Maximum Rating	Safety Margin	Result

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MOSFET V_DS	12.7 V	12.7 V	30.0 V	57.6%	**PASS**
MOSFET Peak Current	0.25 A	0.24 A	5.7 A	95.7%	**PASS**
MOSFET Power Loss	0.8 mW	0.9 mW	1400 mW	99.9%	**PASS**
12V Output Ripple	15 mV	16 mV	120 mV	86.6%	**PASS**
3.3V LDO Output	3.30 V	3.31 V	3.60 V	8.0%	**PASS**